CH10.1 和、倍角關係與弧度量

1. $\dfrac{-8+3\sqrt{5}}{15}$ 2. 3 3. $\dfrac{56}{65}$ 4. $\dfrac{-24-10\sqrt{5}}{25}$ 5. $\dfrac{1}{2}$ 6. 120 7. 25: 39: 16 8. $\dfrac{4\pi}{3}$ 9. 21 10. (24,6)

解析

1. $\because \alpha$ 為第三象限角，且 $\sin\alpha=-\frac{3}{5}$ $\therefore \cos\alpha=-\frac{4}{5}$

又 β 為第四象限角，且 $\cos\beta=\frac{2}{3}$ $\therefore \sin\beta=-\frac{\sqrt{5}}{3}$

$$\cos(\alpha-\beta)=\cos\alpha\cos\beta+\sin\alpha\sin\beta$$

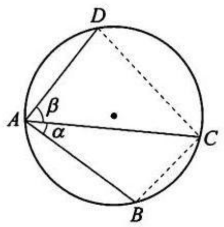
$$=\left(-\frac{4}{5}\right)\times\frac{2}{3}+\left(-\frac{3}{5}\right)\left(-\frac{\sqrt{5}}{3}\right)=\frac{-8+3\sqrt{5}}{15}$$

2. $\tan\alpha\cdot\tan\beta$ 為 $x^2-4x+a=0$ 之兩根

$\therefore \tan\alpha+\tan\beta=4$ 且 $\tan\alpha\tan\beta=a$

$$\tan(\alpha+\beta)=\frac{\tan\alpha+\tan\beta}{1-\tan\alpha\tan\beta}\Rightarrow -2=\frac{4}{1-a}$$

$$\therefore a=\underline{3}$$



3. 如右圖 $\because \dfrac{\overline{BC}}{\sin\alpha}=2R=1$

$$\therefore \sin\alpha=\overline{BC}=\frac{5}{13}$$

又 $\dfrac{\overline{CD}}{\sin\beta}=2R=1$

$$\therefore \sin\beta=\overline{CD}=\frac{3}{5}$$

$\because \alpha,\beta$ 皆為銳角 $\therefore \cos\alpha = \frac{12}{13}, \cos\beta = \frac{4}{5}$

$$\therefore \sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

$$= \frac{5}{13}\times\frac{4}{5} + \frac{12}{13}\times\frac{3}{5} = \underline{\frac{56}{65}}$$

4. $\because 270^\circ < \theta < 360^\circ$ 且 $\cos\theta = \frac{3}{5}$ $\therefore \sin\theta = -\frac{4}{5}$

$$\sin2\theta = 2\sin\theta\cos\theta = 2\times\left(-\frac{4}{5}\right)\times\frac{3}{5} = -\frac{24}{25}$$

又 $135^\circ < \frac{\theta}{2} < 180^\circ$

$$\therefore \cos\frac{\theta}{2} = -\sqrt{\frac{1+\cos\theta}{2}} = -\sqrt{\frac{1+\frac{3}{5}}{2}} = -\frac{2\sqrt{5}}{5}$$

$$\therefore \sin2\theta + \cos\frac{\theta}{2} = \frac{-24-10\sqrt{5}}{25}$$

5. $\because \tan2\theta = \tan(180^\circ - \angle C) = -\tan C = \frac{4}{3}$

$$\therefore \tan2\theta = \frac{2\tan\theta}{1-\tan^2\theta} = \frac{4}{3}$$

$$\Rightarrow 2\tan^2\theta + 3\tan\theta - 2 = 0 \Rightarrow (2\tan\theta - 1)(\tan\theta + 2) = 0$$

$\therefore \tan\theta = \frac{1}{2}$ 或 $\tan\theta = -2$

$$\because \tan C < 0 \ \therefore 90^\circ < C < 180^\circ$$

即 $0^\circ < \theta < 90^\circ$ $\therefore \tan\theta = \frac{1}{2}$

6. $\because 0^\circ < \angle A, \angle B < 90^\circ$

$$\therefore \cos A = \sqrt{1-\sin^2 A} = \sqrt{1-\left(\frac{13}{14}\right)^2} = \frac{3\sqrt{3}}{14}$$

$$\cos B=\sqrt{1-\sin^2 B}=\sqrt{1-\left(\frac{11}{14}\right)^2}=\frac{5\sqrt{3}}{14}$$

$$\because 0^\circ<\angle A+\angle B<180^\circ$$

$$\therefore \cos(A+B)=\cos A\cos B-\sin A\sin B$$

$$=\frac{3\sqrt{3}}{14}\times\frac{5\sqrt{3}}{14}-\frac{13}{14}\times\frac{11}{14}=-\frac{98}{196}=-\frac{1}{2}$$

即 $\angle A+\angle B=\underline{120^\circ}$

7. $\because \cos B=-\frac{4}{5}<0$

$\therefore \angle B$ 為鈍角，即 $\angle A$ 、 $\angle C$ 皆為銳角

又 $\sin A=\frac{5}{13}\Rightarrow \cos A=\frac{12}{13},\sin B=\frac{3}{5}$

$$\sin C=\sin[180^\circ-(\angle A+\angle B)]=\sin(A+B)$$

$$=\sin A\cos B+\cos A\sin B$$

$$=\frac{5}{13}\times\left(-\frac{4}{5}\right)+\frac{12}{13}\times\frac{3}{5}=\frac{16}{65}$$

$$\therefore a:b:c=\sin A:\sin B:\sin C$$

$$=\frac{5}{13}:\frac{3}{5}:\frac{16}{65}$$

$$=\underline{25:39:16}$$

8. 時針移動 $30^\circ=\frac{\pi}{6}$ (弧度)

所求 $=8\times\frac{\pi}{6}=\frac{4\pi}{3}$ (公分)

9. 令 $\angle AOB=\theta,4=2\theta\Rightarrow\theta=2$ (弳)

所求 = 扇形 OAB 面積-扇形 OCD 面積

$$=\frac{1}{2}\times 5^2\times 2-\frac{1}{2}\times 2^2\times 2=25-4=\underline{21}$$

10. 設扇形的半徑 r ,弧長 s

則 $\frac{1}{2}rs = 36 \Rightarrow rs = 72$

周長為 $2r + s$

由算幾不等式知 $\frac{2r+s}{2} \geq \sqrt{(2r)s} = 12$

$\Rightarrow 2r + s \geq 24$,即 $L = 24$

「$=$」成立 $\Leftrightarrow 2r = s = 12 \Rightarrow r = 6$,即 $r_0 = 6$

故數對 $(L, r_0) = (24,6)$